

- 5 (a) Fig. 5.1 shows four types of cell that can be seen in a prepared slide of blood taken from a mammal.

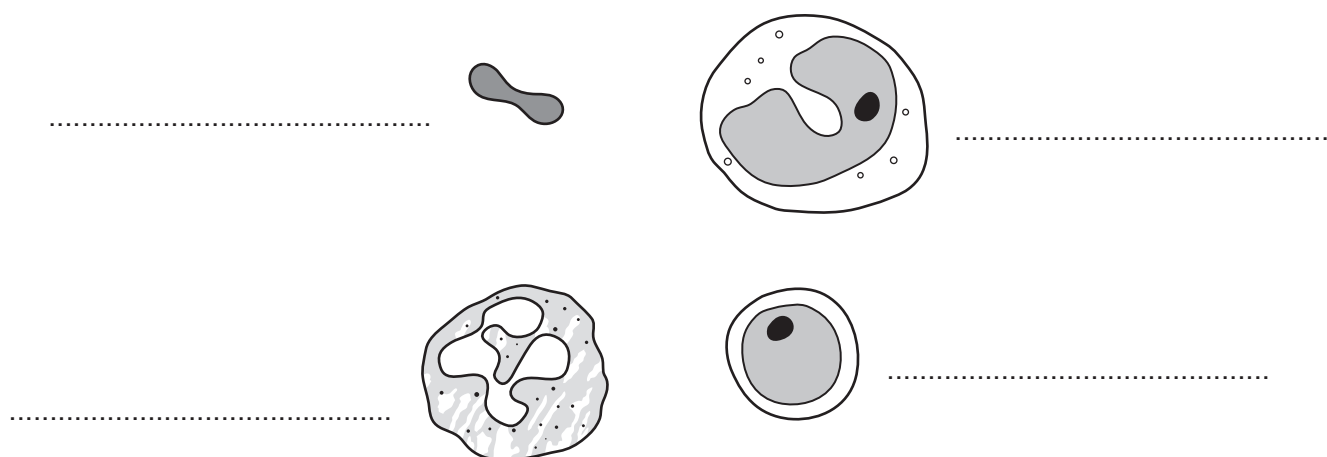


Fig. 5.1

Identify the four types of cell shown in Fig. 5.1.

Write the name of the cell type on the answer line provided by each cell in Fig. 5.1.

[4]

- (b) During systole and diastole of the cardiac cycle, changes in blood pressure occur in the four chambers of the heart.

Fig. 5.2 shows changes in the blood pressure in the **left** side of the heart and the aorta during one cardiac cycle.

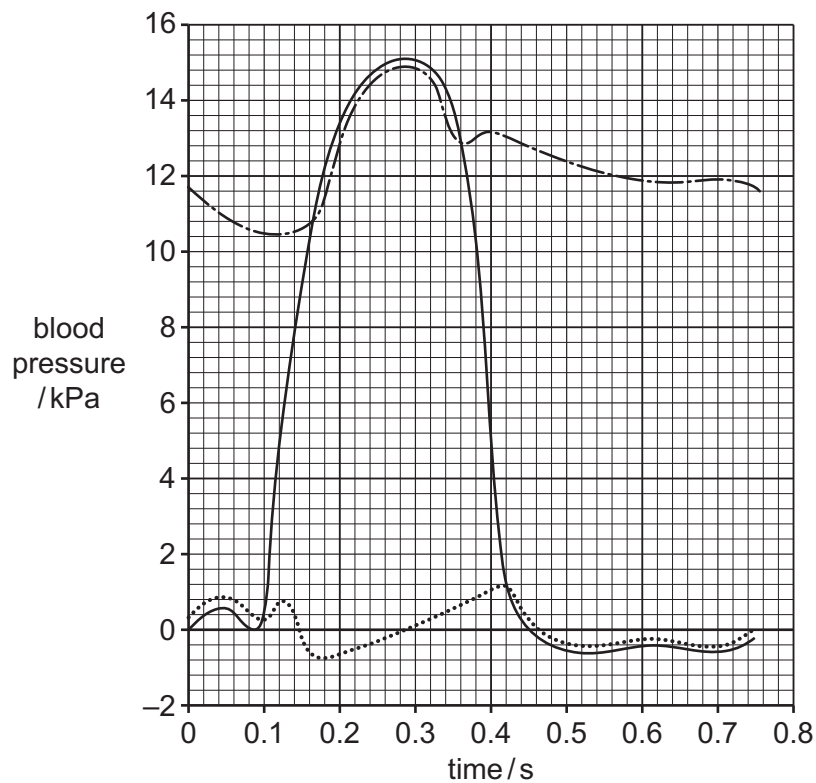


Fig. 5.2

With reference to the blood pressure changes shown in Fig. 5.2:

- state the maximum blood pressure reached in the left ventricle kPa
- state the time at which the bicuspid (left atrioventricular) valve closes. s

- (c) Chronic obstructive pulmonary disease (COPD) can cause a condition known as pulmonary hypertension. This involves an increase in systolic blood pressure in the right ventricle and in the pulmonary arteries.

Fig. 5.3 is a summary of some of the events that can result from COPD.

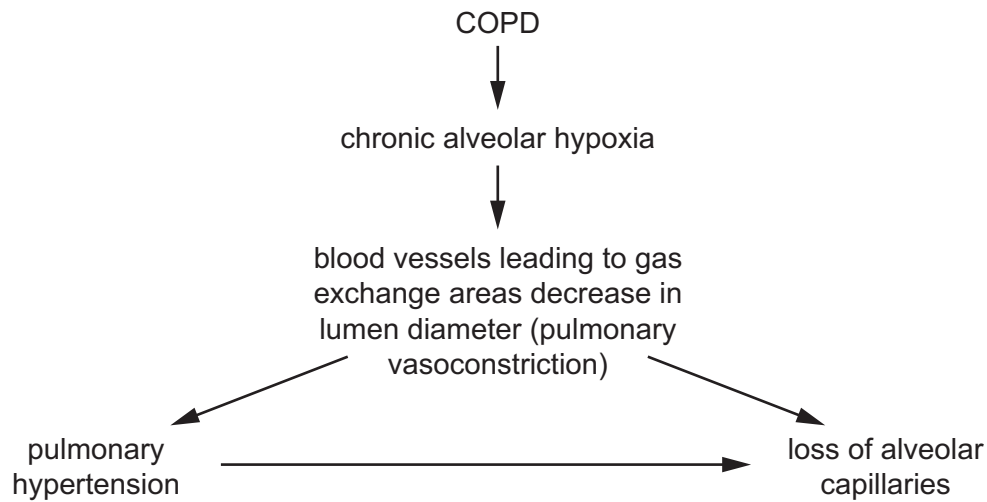


Fig. 5.3

- (i) Chronic alveolar hypoxia describes a condition where the partial pressure of oxygen in the gas exchange regions of the lungs is always lower than normal.

Explain how COPD leads to chronic alveolar hypoxia.

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- (ii) Explain how the loss of alveolar capillaries affects the functioning of the lungs in a person with COPD.

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